

7(a)	<table><tr><th>compound</th><th>location</th></tr><tr><td>PEP carboxylase</td><td>C</td></tr><tr><td>RuBP</td><td>B</td></tr><tr><td>rubisco</td><td>B</td></tr></table>	compound	location	PEP carboxylase	C	RuBP	B	rubisco	B	3
compound	location									
PEP carboxylase	C									
RuBP	B									
rubisco	B									
7(b)	any 2 of: 1 stop, air / oxygen, getting to cells B / rubisco / RuBP / bundle sheath cells ; 2 so rubisco does not catalyse reaction of oxygen with RuBP ; 3 reduce / stop, photorespiration ;	2								
7(c)	any 3 of: describe 1 (enzyme in) <i>Amaranthus</i> / C4, peaks, at higher temperature or (enzyme in) <i>Amaranthus</i> / C4, has higher optimum temperature or (enzyme in) <i>Pisum</i> / C3, has higher activity at low(er) temperatures ; ora 2 data quote ; e.g. C4 – 45 °C and C3 – 30 °C at 100% or C4 – 70% and C3 – 25% at 50 °C ; explain 3 enzyme in, <i>Pisum</i> / C3, less thermostable / denatures at lower temperature ; ora 4 AVP ; e.g. <i>Amaranthus</i> adapted to a hot climate / ora e.g. enzyme in <i>Amaranthus</i> has more S-S bonds	3								